

Detection and Tracking in VisDrone2019-MOT

Detector-limited failures, camera motion, and small-object tradeoffs

Computer Vision MOT Project

VisDrone2019-MOT validation split

June 7, 2026

Modular detection + tracking experiments with MOT-format evaluation.

Problem and Dataset

Drone MOT challenges

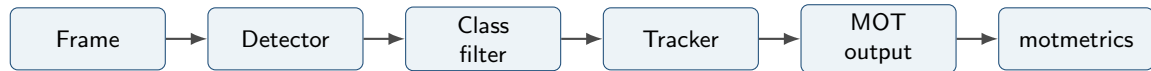
- Small cars and pedestrians at low pixel area
- Occlusion and dense road/intersection scenes
- Moving camera breaks image-space motion assumptions
- Blur and viewpoint changes reduce detector recall

Evaluation scope

- Dataset: VisDrone2019-MOT-val
- Classes: pedestrian and car
- MOT metrics: MOTA, IDF1, IDS, FPS
- Detection diagnosis: precision, recall, AP50

Main question: are failures caused by missed detections, bad associations, false positives, or speed tradeoffs?

Modular Pipeline



Design choice

- Detector and tracker are decoupled
- det/ detections are ignored
- Outputs are saved in MOT format

Why this matters

- Detection-only diagnosis before tracking
- Tracker ablations without changing detector
- Per-sequence failure analysis

Methods Compared

Group	Method	Role
M1	YOLOv11 + SORT	Geometric baseline: Kalman prediction + IoU assignment
M2	YOLO26 + ByteTrack	Fast confidence-aware baseline with low-confidence recovery
M3	RT-DETR + BoT-SORT	Accuracy-oriented detector with appearance cues and GMC
M4	SAHI / upscaling variants	Small-object strategies for low-resolution targets

The experiments are compact: each method probes a specific failure mode instead of expanding a blind grid.

YOLO26 vs RT-DETR

	YOLO26	RT-DETR
Detector type	One-stage dense detector	Transformer detector
Prediction style	Local, grid/anchor-like dense predictions	Query-based global reasoning
Strength	High FPS; strong baseline precision	Better context for crowded aerial scenes
Weakness	Small objects can be missed	Slower; threshold sensitive
MOT implication	Good speed and usable MOTA	Higher recall can help, but FP control is critical

RT-DETR is attractive for aerial scenes because global context can recover small or ambiguous cars, but it must be calibrated for tracking.

Tracking: Geometry vs Appearance

Tracker	Association signal	Failure pressure
SORT	Kalman motion + IoU	Camera motion, occlusion, similar nearby targets
ByteTrack	High/low confidence detections + IoU	Better recall use, still mostly geometric
BoT-SORT	Motion + IoU + appearance/ReID + GMC	More robust IDs, slower and detector-dependent

Interpretation

Pure geometry is fragile when the camera moves or objects overlap. Appearance embeddings and global motion compensation add association cues that reduce identity fragmentation.

Baseline Run: First Diagnosis

Method	MOTA	IDF1	IDS	FP	FN	FPS
M1 YOLOv11 + SORT	0.137	0.336	515	11.9k	43.0k	40.3
M2 YOLO26 + ByteTrack	0.169	0.341	384	8.8k	44.2k	56.5
M3 RT-DETR + BoT-SORT	0.214	0.436	102	8.8k	41.5k	4.6
M4 SAHI + ByteTrack	0.006	0.377	668	29.8k	33.4k	3.9

Initial read

- M2 was the fast practical baseline.
- M3 reduced FN and IDS, showing useful association.

Problem

- RT-DETR was threshold-sensitive.
- SAHI recall gains produced too many false tracks.

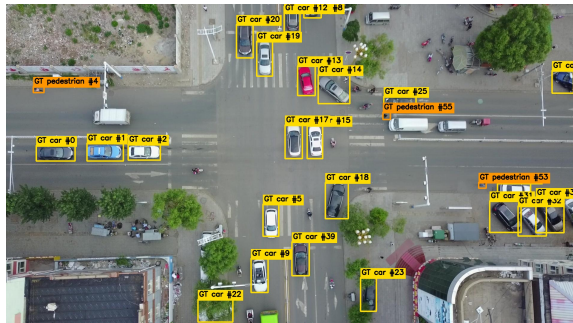
Failure mode: high recall can hurt MOTA if false positives explode.

Crossroad Failure: YOLO26 vs Ground Truth

YOLO26 baseline



Ground truth



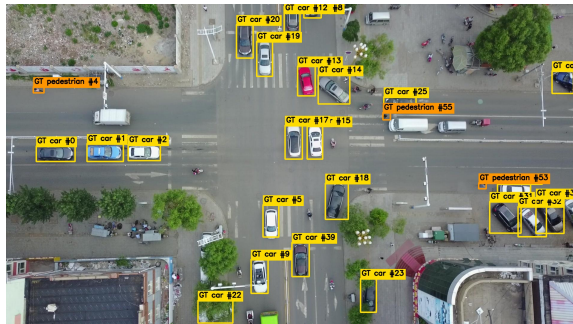
Many small cars are never detected, so the tracker cannot recover them.

Crossroad Improvement: RT-DETR + BoT-SORT + GMC

RT-DETR conf055 + GMC



Ground truth



RT-DETR improves car coverage, but many tiny objects remain missed.

RT-DETR Confidence Tuning

RT-DETR conf	MOTA	IDF1	IDS	FP	FN	FPS
0.50	0.211	0.465	133	11.6k	38.9k	4.0
0.55	0.214	0.436	102	8.8k	41.5k	4.2
0.60	0.206	0.402	99	6.6k	44.2k	4.0
0.65	0.187	0.365	95	4.8k	47.3k	3.9

Selection rule

Conf 0.55 is the best full-validation MOTA; conf 0.50 is more recall/IDF1-oriented.

Crossroad: Global Best vs Sequence-Oriented Variant

RT-DETR conf055 + GMC



RT-DETR conf050



Conf0.50 is more recall-oriented and was stronger on this scene, but conf0.55 was selected globally because it controls FP and IDS better.

GMC Ablation: Camera Motion

Family	GMC	MOTA	IDF1	IDS	FPS
RT-DETR + BoT-SORT	off	0.206	0.391	294	4.22
RT-DETR + BoT-SORT	on	0.214	0.436	102	4.20
YOLO26 + BoT-SORT	off	0.171	0.332	323	53.97
YOLO26 + BoT-SORT	on	0.180	0.370	176	38.98

Effect

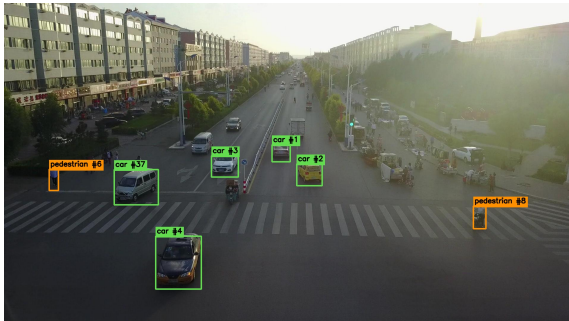
- MOTA improves by about +0.008.
- IDF1 improves by about +0.04.
- IDS drops by 192 for RT-DETR and 147 for YOLO26.

Exception

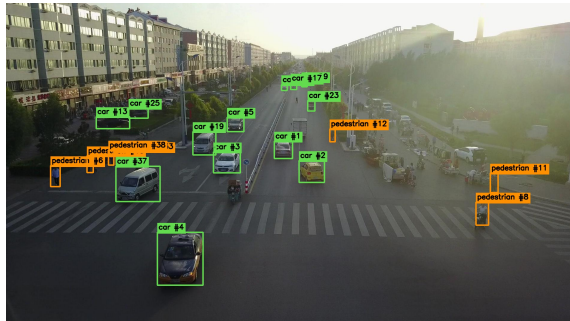
- uav0000182_00000_v was the notable sequence-level metric exception.

Camera Motion: GMC Off vs On

GMC off



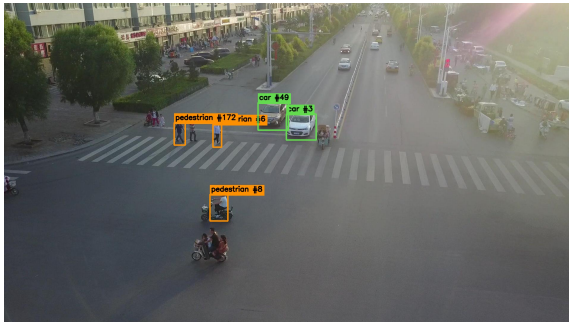
GMC on



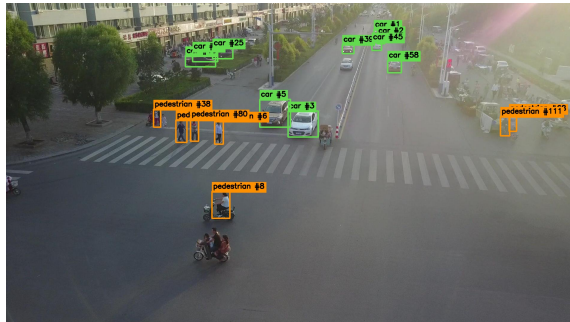
GMC stabilizes association when global camera motion shifts object positions. IDS 294 to 102;
IDF1 0.391 to 0.436.

GMC: Identity Stability Under Camera Motion

GMC off



GMC on



The gain is not more detections; it is fewer ID switches and more stable tracks. MOTA 0.206 to 0.214.

Detection Diagnosis: uav0000305_00000_v

Detector	Class	Precision	Recall	AP50	FP	FN
YOLO26	car	0.814	0.160	0.141	132	3040
RT-DETR conf055	car	0.890	0.314	0.298	141	2482
YOLO26	pedestrian	0.000	0.000	0.000	0	540
RT-DETR conf055	pedestrian	0.000	0.000	0.000	8	540

Diagnosis

This sequence is detector-limited: RT-DETR nearly doubles car recall, but most cars are still missed. Pedestrian recall is zero for both detectors.

Best MOT on this sequence came from RT-DETR conf050, but detection recall remains the bottleneck.

Crossroad: Detector Choice Matters

YOLO26 baseline



RT-DETR conf055 + GMC



RT-DETR provides visibly better coverage than the YOLO26 baseline on this aerial view.

Crossroad: SAHI Small-Object Tradeoff

RT-DETR conf055 + GMC



SAHI strict



SAHI can reveal more small objects, but it also introduces clutter and false tracks.

SAHI and Upscaling: Small-Object Tradeoff

Method	Car recall proxy	MOTA	IDF1	FP	FN	IDS
YOLO26 baseline	0.299	0.169	0.325	6.0k	47.0k	332
Upscale 2.0 + BoT-SORT	0.499	0.110	0.463	23.8k	33.1k	295
SAHI 768/0.15/conf0.40	0.473	0.092	0.423	22.5k	35.5k	281
RT-DETR conf055 + GMC	0.294	0.214	0.436	8.8k	41.5k	102

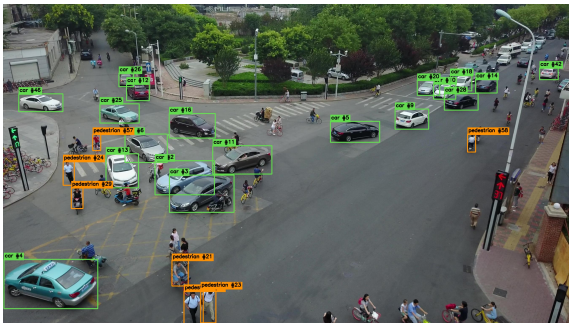
Readout

SAHI and upscaling recover more small objects, but the added detections increase FP and identity instability.

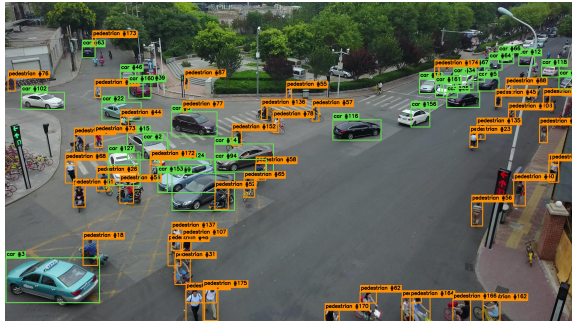
Recall alone is not enough for MOT; precision and identity stability decide whether detections help.

SAHI: More Detections, More Risk

RT-DETR conf055 + GMC



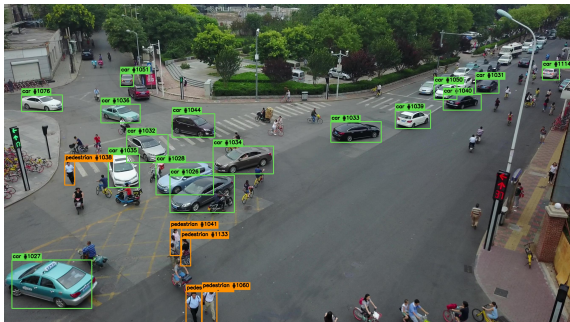
SAHI strict



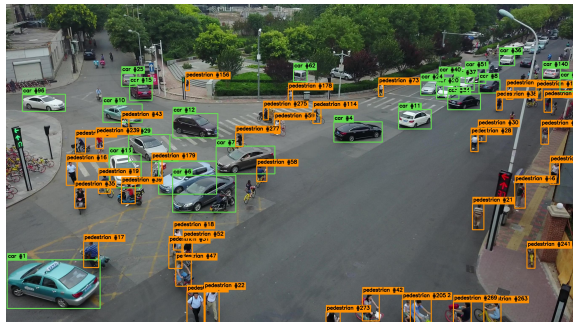
For MOT, extra detections only help if they do not become false tracks.

Upscaling: Recall-Oriented Small-Object Mode

YOLO26 baseline



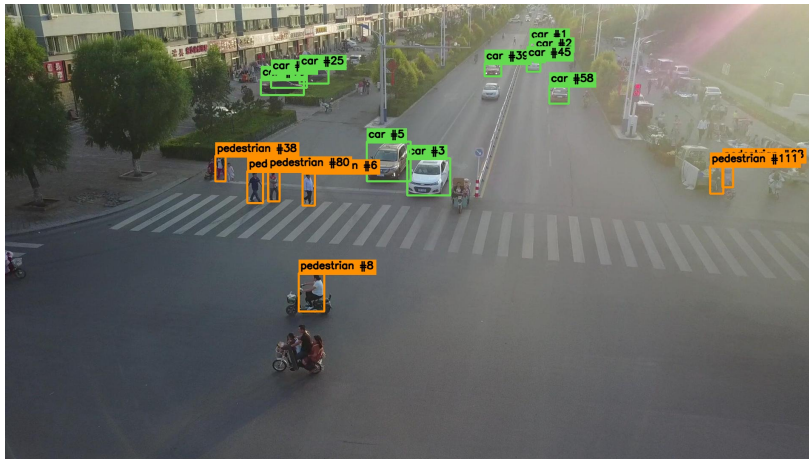
YOLO26 upscale 2.0 + BoT-SORT



Upscaling increased the car recall proxy, but also increased FP and reduced MOTA.

Final Method: Stable Tracking Example

RT-DETR conf055 + GMC



The selected method balances detection quality with robust association under drone motion.

Final Conclusion

Selected method

RT-DETR + BoT-SORT + GMC,
conf=0.55

Best full-validation MOTA, very low IDS, strong IDF1, and explicit camera-motion handling.

Tradeoff

- Slower than YOLO26 methods.
- Still detector-limited on crossroad scenes.

Lessons

- Aerial crossroads are dominated by missed cars.
- Appearance + GMC improves identity stability.
- SAHI/upscaling need stricter FP control.

Next steps

- Fine-tune detector on VisDrone classes.
- Improve small-object postprocessing.
- Add sequence-adaptive detector/tracker settings.